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Research on Phenomenology of HNL and Possible Signals in Multi-Messenger Observation

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Recently, the IceCube detector has identified AGN NGC 1068 as a point source of high-energy neutrinos, while the γ ray observation exceeds the prediction of electromagnetic cascade models. This suggests some of the γ ray energy can be transported outside from being attenuated. In this paper we propose HNL, Heavy Neutral Leptons, as a way to explain this observation. We first review the origin of HNL theory, calculating the phenomenology behavior of certain processes related to $p - \gamma$ and $p - p$ neutrino production, then use it to try to explain the γ ray observation of AGN NGC 1068, and several other astronomical observations.

Keywords: HNL; multi-messenger; NGC 1068

1. Introduction

As advancements are made in astronomical observations, different types of new signals have become observable, along side with gravitational waves and wider range of electro-magnetic signals, neutrino signals are now observed by IceCube, which has recently spotted AGN NGC 1068 as a point source of neutrino flux around energy of 1 TeV,¹ and several other experiments, such as KM3NeT/ARCA, while still under construction, has observed a neutrino signal which possibly has energy as high as 220 PeV.²

Further more, neutrino has always been the strange one in Standard Model (SM), because of its tiny mass, thus weak coupling with the Higgs field, is considered some what unnatural. This leads neutrino and the origin of its small mass to be considered as a key to Beyond Standard Model Physics (BSM), such as Heavy Neutral Lepton (HNL) being candidate of dark mater. A lot of experiments on the earth is working on finding possible signals coming from BSM theories.³ Although no definite signal has been observed, they do provide a solid constrain on the parameter space of possible BSM theories, for HNL with $3 \text{ MeV} < m_N < 19.5 \text{ MeV}$, the mixing angle

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with muon neutrino is constrained roughly as $|U_{\mu N}|^2 \lesssim 10^{-4}$.⁴

And, in the observation of NGC 1068 done by IceCube, when comparing with theoretical predictions⁵ based on corona cascade models, exceeds the prediction of observed γ -ray flux around 10 GeV energy. This suggests that the hidden core of NGC 1068 has some process to transport more, electro-magnetic energy out. Axion is one option,⁶ another option is HNL, which is discussed in this paper. Before NGC 1068, there are already attempts to combine multi-messenger observations in favor of constraining BSM theories.⁷⁻⁹

This paper is structured as follows. In Section , we outline the basic of HNL theory originating from neutrino mass, and calculate its phenomenology properties, while taking polarization into account. In Section , we investigate the multi-messenger observation results of NGC 1068, try to explain the exceeding γ -ray flux. Conclusion is in Section .

2. ISS Model

To explain the tiny mass of neutrino, a bunch of models known as Seesaw Mechanisms are proposed. For Type-I Seesaw model, we have

$$m_\nu = \frac{m_D^2}{M_R}, \quad m_N = M_R, \quad |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \quad (1)$$

Where m_D is the Dirac mass of neutrino, and M_R is the Majorana mass of the added right-handed neutrino field. For a sufficiently natural m_D and m_R , the mixing angle $|U|^2$ is always too small which makes observation impossible. How every by introducing another fermionic field, the mass matrix becomes

$$(\overline{\nu_L} \quad \overline{\nu_R^c} \quad \overline{S^c}) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \quad (2)$$

The new parameter μ stands for potential lepton number violation, which should be small. The basic properties of neutrino and a pseudo-dirac fermion pair, which can be treated as a fermion because of the small μ , are

$$m_\nu = \mu \frac{m_D^2}{M^2}, \quad m_N = M, \quad |U|^2 = \frac{m_\nu}{\mu} \quad (3)$$

3. HNL Production

Such HNL particles only interact through its mixed part of SM neutrino, which means those process which produces SM neutrino, if not prohibited by phase space, will also produce HNL at the same time.

3.1. Pion decay

Under the condition of AGN, $p-p$ and $p-\gamma$ processes are though to be the origin of high energy cosmic rays.¹⁰ Those processes creates a large number of pions, and the

decay of high-energy pions turns into high-energy cosmic rays, including neutrinos, γ -rays.¹¹

After introducing HNL particle mixing with SM neutrino, charged pions can decay into HNL particles, intermediated by W bosons. By slightly modifying the

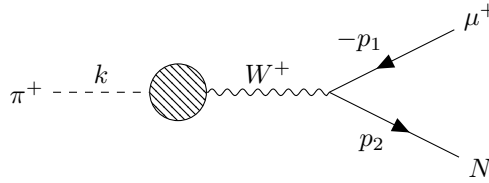


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNL particle.

process of SM pion decay, we find that the decay width to HNL is

$$\frac{\Gamma_{\pi \rightarrow \mu N}}{\Gamma_{\pi \rightarrow \mu \nu}} = |U_\mu|^2 \times \rho_\pi(m_N) \quad (4)$$

Where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2 + m_N^2) - (m_\mu^2 - m_N^2)^2) \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{m_\mu^2(m_\pi^2 - m_\mu^2)^2} \quad (5)$$

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx \quad (6)$$

It is useful to determine the helicity of the HNL particle, since it will play a role in the angular dependence when working on the decay of HNL. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2 + m_N^2)m_\pi^2 - (m_\mu^2 - m_N^2)^2 - s(m_\mu^2 - m_N^2) \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)} \quad (7)$$

Where s is the helicity of HNL, and $s = +1$ for right-handed polarized HNL particle.

There are no differences in angular dependence for different helicity end states, since pion is unpolarized. After boosting to the lab frame, we must account for the polarization change on the HNL particles, thus for a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J\left(\frac{E_N}{x}\right) |U_\mu|^2 \rho_\pi \times r_{(s)}(x, E_N) \times \frac{dx}{(\lambda^+ - \lambda^-) \cdot x} \quad (8)$$

Where

$$\lambda^\pm = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{2m_\pi^2} \quad (9)$$

$$\cos \theta_r = \frac{x(m_\pi^2 - m_\mu^2 + m_N^2) - 2m_N^2}{x \sqrt{(1 - \frac{m_N^2}{E_N^2}) \lambda(m_\pi^2, m_\mu^2, m_N^2)}} \quad (10)$$

$$r_{(s)}(x, E_N) = \frac{\Gamma_{(s)} \frac{1 + \cos \theta_r}{2} + \Gamma_{(-s)} \frac{1 - \cos \theta_r}{2}}{\Gamma_{(s)} + \Gamma_{(-s)}} \quad (11)$$

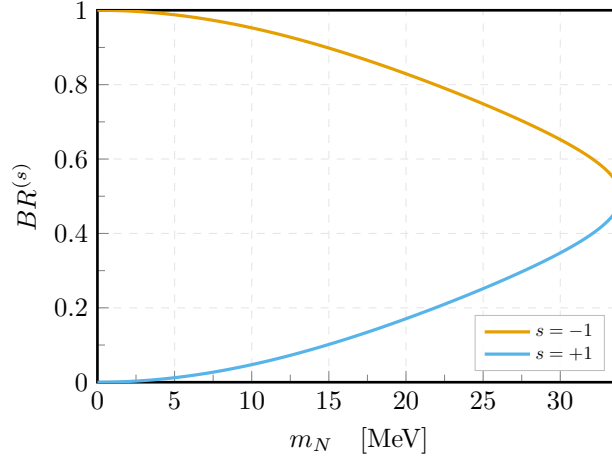
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Fig. 2. Eq. 7 for different s . In pion decay, right-handed HNLs are more likely to be produced when near the edge of phase space.

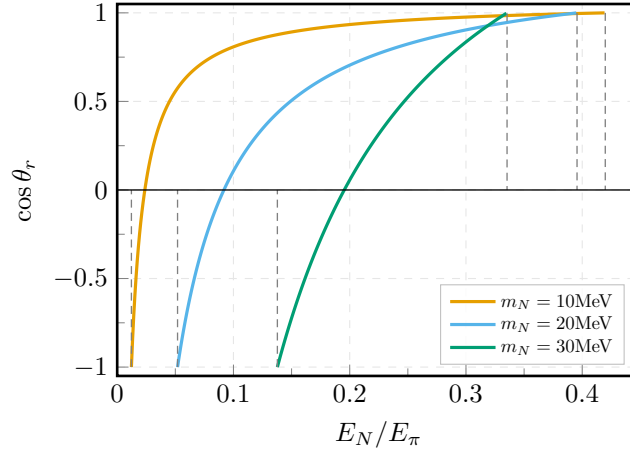


Fig. 3. Eq. 10 visualized with different m_N values. Eq. 10 determines how much helicity flip will happen during Lorentz boost. While when m_N is small $\cos\theta_r \approx 1$, combined with Fig. 2, HNLs produced are almost left-handed; as m_N approaches the edge of phase space, helicity flip is more significant and right-handed HNLs cannot be ignored.

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,¹¹ with modifications come from HNL mass and helicity separation.

3.2. Muon decay

Muons will decay into HNL particles through a similar process as they decay into ν_μ . We find the decay width of $\mu^- \rightarrow N e^- \bar{\nu}_e$ has the same form as the process of τ decay which has been studied,¹² since this process is symmetric between μ and τ ,

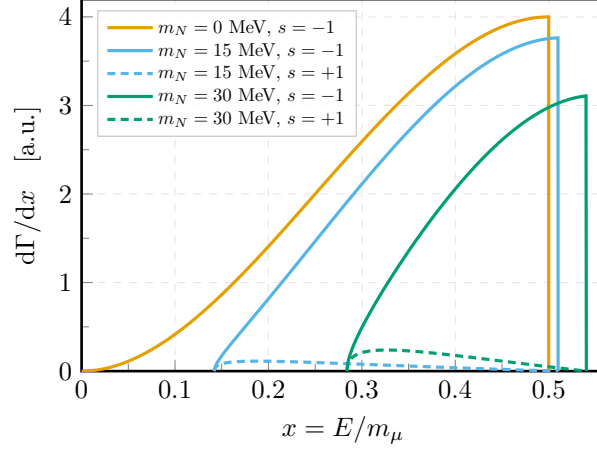


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in muon rest frame. Normalized under $m_N = 0$, which basically resembles SM neutrino, with no right-handed product exists. When m_N increases, shape of the spectrum stays roughly the same, but the max energy increases while the total decay width decreases, and more right-handed HNLs, though still a small fraction, will be produced.

which is

$$\frac{\Gamma_{\mu \rightarrow N}}{\Gamma_{\mu \rightarrow \nu_\mu}} = |U_\mu|^2 \times \rho_\mu(m_N) \quad (12)$$

Where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4 \ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \quad (13)$$

And the spectrum of HNL particle under muon rest frame, separated by polarization, is

$$\begin{aligned} \frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3} |U_\mu|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu \\ - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \end{aligned} \quad (14)$$

Several examples are plotted in Fig. 4.

Though one can keep the muon polarization and calculate the angular dependence of how many different helicity HNL particles will be produced while the angle respect to muon polarization changes, we will simply ignore the muon polarization here as we assume that the magnetic environment of AGN will wipe the polarization out.

For a given muon spectrum, we have

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$$Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y} Q_\mu\left(\frac{E_N}{y}\right) |U_\mu|^2 \rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \cdot \left(\frac{1 + \cos \theta_r}{2} g_N^{(s)}(x) + \frac{1 - \cos \theta_r}{2} g_N^{(-s)}(x) \right) \quad (15)$$

Where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2}, \quad x^- = \frac{1}{2} \left(y + \frac{1}{y} \cdot \frac{m_N^2}{m_\mu^2} \right) \quad (16)$$

$$\cos \theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \quad (17)$$

$$\cos \theta_r = \frac{1}{y \sqrt{1 - \frac{m_N^2}{E_N^2}}} \left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x \cos \theta_0 \right) \quad (18)$$

While $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N^2}{m_\mu^2}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \quad (19)$$

4. HNL Decay

Straight-forwardly, HNL particles under ISS model will , and will only, decay into other leptons. Possible products are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes are well studied,¹² we can recall the decay width for each process as

$$\Gamma_{N \rightarrow 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \quad (20)$$

$$\Gamma_{N \rightarrow \nu e \bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \times C_1^e \quad (21)$$

Where

$$C_1^e = \frac{1}{4} (1 - 4 \sin^2 \theta_W + 8 \sin^4 \theta_W) \quad (22)$$

To simplify the problem we will ignore electron mass, and 3ν end state has taken all different flavor combinations into account. Thus the branch ratio of HNL particle processing a visible decay is

$$BR_{N \rightarrow \nu e \bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \quad (23)$$

The energy loss is pretty significant. Only one tenth of HNL decays will be observable. The angular and energy distributions of electrons/positrons are

$$g_e^{(1)}(x, \cos \theta) = \frac{8}{C_1^e} \cdot x^2 \cdot \{[(4 - 16s_W^2 + 64s_W^4)x - (1 - 4s_W^2 + 28s_W^4)] \cos \theta - (4 - 16s_W^2 + 64s_W^4)x + (3 - 12s_W^2 + 36s_W^4)\} \quad (24)$$

$$g_e^{(2)}(x, \cos \theta) = \frac{16}{C_1^e} \cdot x^2 \cdot \{[(6 - 24s_W^2 + 32s_W^4)x - (3 - 12s_W^2 + 14s_W^4)] \cos \theta - (6 - 24s_W^2 + 32s_W^4)x + (3 - 12s_W^2 + 18s_W^4)\} \quad (25)$$

Where $x = E_e/m_N$, $s_W = \sin \theta_W$. θ is the angle between the out-going electron and initial HNL spin.

To get the total electron spectrum from a given HNL spectrum, we integral over the HNL spectrum and get

$$Q_e^{(Lab)}(E) = BR_{N \rightarrow \nu e \bar{e}} \cdot \int_E^{+\infty} dE_N Q_N^{(s)}(E_N) \cdot \int_{\frac{E}{2E_N}}^{\frac{1}{2}} \frac{dx}{xE_N} \sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N} - 1)) \quad (26)$$

Combining with 24 and 25, the remaining integral is

$$Q_e^{(Lab)}(E) = BR_{N \rightarrow \nu e \bar{e}} \cdot \int_0^1 \frac{dy}{3y} Q_N^{(s)}(\frac{E}{y}) [16y^3 - 27y^2 + 11 - s \cdot (32y^3 - 63y^2 + 36y - 5)] \quad (27)$$

Where $y = E/E_N$, and θ_W no-longer appears in this formula.

5. HNL in NGC 1068

According to observation of IceCube, NGC 1068 has been spotted as a neutrino point source, and the major way of producing neutrinos is through pion decays.⁵ By combining the analytical results of neutrino spectrum from pion decays¹¹ and the theoretical prediction of neutrino spectrum of NGC 1068,⁵ We use the following fit as a approximation of the pion spectrum in NGC 1068

$$\Phi_{\pi^\pm}(E) = \Phi_0 \cdot E^{-\Gamma} \cdot e^{-(E/E_c)^\beta} \quad (28)$$

Where the best fit, which is visualized in Fig. 5, is

$$\Phi_0 = 10^{-11.3} \text{ GeV}^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}, \Gamma = 0.1, E_c = 0.8 \text{ GeV}, \beta = 0.3 \quad (29)$$

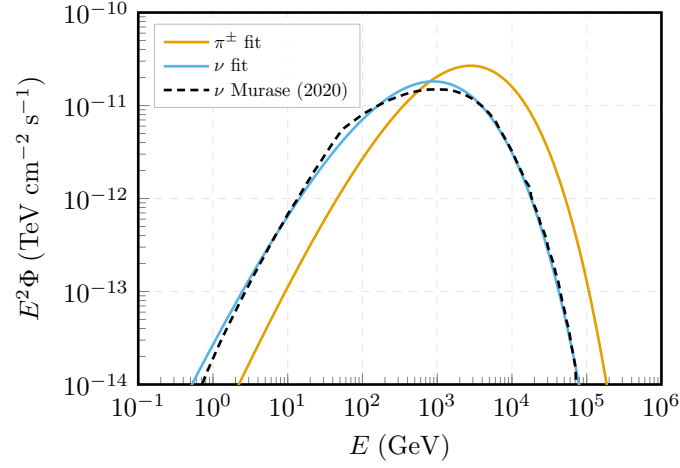


Fig. 5. Show case of the simple parameterization of NGC 1068 pion spectrum in Eq. 28 for theoretical prediction.⁵

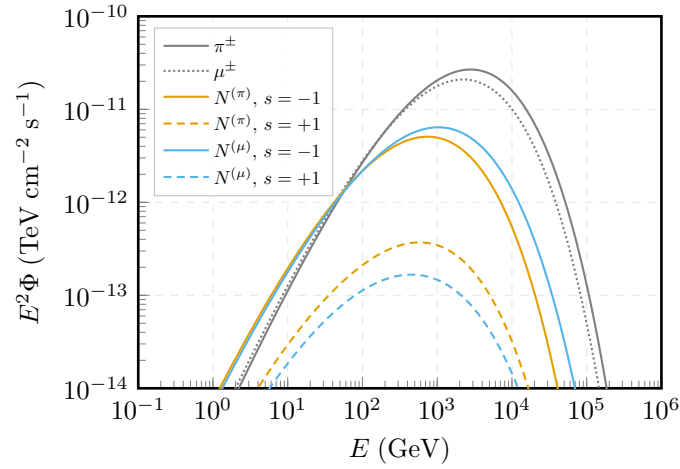


Fig. 6. HNL spectrum from pion and muon decay for NGC 1068, using Eq. 28 as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10$ MeV, and all HNL spectrums are scaled assuming $|U_\mu|^2 = 1$ to better show the physics.

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